

MISO WEST AFFECTED SYSTEM IMPACT STUDIES

MISO DPP-2017-FEB-WEST

SOUTHWEST POWER POOL, INC.

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Document Revisions

NO.	Revision	Date	PRD	CHK	APV
1	Draft Report Issued for Review	2/26/2021	WB	WB	WB
2	Revision 1	3/3/2021	WB	WB	WB
3	Revision 2	3/4/2021	WB	WB	WB
4	Final Report	3/10/2021	WB	WB	WB

Introduction

MISO requested a definitive Affected System Impact Restudy from SPP for the DPP-2017-FEB-West Phase 3 cluster. The purpose of this analysis was to determine the impact of the MISO generation interconnection requests on the SPP transmission system. Additionally, the analysis looked to identify the amount of Interconnection Service available to the projects resulting in no constraints requiring mitigation. This analysis evaluated two MISO interconnection requests in SPP cluster groups 9 and 15 with a total generation capacity of 249.48 MW. While results from this analysis will be considered final, a restudy may be required should significant changes to the study assumptions occur¹. The generation interconnection requests analyzed in this Affected System Impact Study are listed in **Appendix A** by queue number, amount, requested interconnection service type, area, and proposed interconnection point.

The Siemens Power Technologies International PSS/E Version 33.11.0 and PowerGEM's TARA 2002 were used for this analysis. SPP provided the following DISIS-2017-001 BASE case models:

- Year 1 (2019) Winter Peak (19WP)
- Year 2 (2020) Spring (20G)
- Year 2 (2020) Summer Peak (20SP)
- Year 5 (2024) Light (24L)
- Year 5 (2024) Summer Peak (24SP)
- Year 5 (2024) Winter Peak (24WP)
- Year 10 (2029) Summer Peak (29SP)

EPE updated power flow cases to reflect the groups under study and developed a total of forty-six (46) cases, specifically 23 Base Cases (BC) and 23 Transfer Cases (TC). The power flow analysis was performed to determine if the transmission system can accommodate the injection from the current study cluster generation interconnection requests without violating SPP's transmission planning criteria outlined below in the Study Methodology Criteria Section.

The Affected System Impact Study (ASIS) has been conducted consistent with the Southwest Power Pool (SPP) Open Access Transmission Tariff (OATT), SPP-MISO Joint Operating Agreement (JOA), and SPP Business Practices to determine definitive impacts to the SPP transmission system.

¹ Significant changes to study assumptions include but are not limited to interconnection request withdrawals and/or changes to higher queued network upgrades included in the base case.

Base Case Model Build and Dispatch

DC Scan

The DC Scan analysis was performed to determine which interconnection requests should be included in the analysis. The DC Scan was also performed to determine how the in-scope interconnection requests should be dispatched (in-group or out-group). **Table 1** lists the Distribution Factor (DF) cut-off criteria used to perform DC Scan analysis. The Distribution Factor (DF) cut-off criteria used for DC Scan is based on SPP's Transmission Distribution Factor (TDF) criteria used to identify constraint and mitigation.

Table 1: DC Scan DF Criteria

Tier	Description	Criteria (ERIS System Intact)	Criteria (ERIS Outage)	Criteria (NRIS System Intact or Outage)
Tier 1	In-scope, In-group	DF $\geq$ 3%	DF $\geq$ 20%	DF $\geq$ 3%
Tier 2	In-scope, Out-group	DF < 3%	DF $\geq$ 3%	DF $\geq$ 3%
Tier 3	Out-scope	DF < 3%	DF < 3%	DF < 3%

The results of DC Scan analysis revealed that both projects from DPP-2017-FEB-West P3 met the DC Screening criteria and hence they both were included in the analysis. **Table 2** provides a summary of DC Scan analysis. Distribution factors were calculated using each project as the “sending” system and the MISO Classic² region as the “sink” system. Detailed results for the DC Scan can be found in **Table 3**.

Table 2: DC Scan Results by Queue Cluster

Queue Cycle	Region	MISO Classic 3%	
		Include	Exclude
Feb-17	West	2	0

Table 3: Detailed DC Scan Results

Results	
DC Scan Detailed Results	 DC_Screen_Details.xlsx

Base Case Model Review and Grouping

SPP provided the following information to EPE:

1. List of interconnection requests for consideration in the analysis.
2. List of all higher queued interconnection requests.
3. DISIS-2017-001 BASE cases.
4. Latest MISO bench and study cases used for the DPP-2017-FEB-West P3.
5. Current study network upgrades previously identified by SPP for consideration in the analysis.

² MISO Classic is defined as the PSSE areas of 207, 208, 210, 216, 217, 218, 219, 295, 296, 314, 333, 356, 357, 360, 361, 600, 608, 613, 615, 620, 627, 633, 635, 661, 680, 694, 696, 697, and 698

The DPP-2017-FEB-West P3 Affected System Impact Study included two (2) generation interconnection requests (“GIRs”) in the MISO footprint. **Appendix A** lists the current study cluster generation interconnection requests included in the study. The BASE cases are ITP cases that have topology added for all queue generators, but all queue generators are not necessarily dispatched. The MISO West study generators, including all available collector system data, were added, and kept offline in BASE models listed below:

- Year 1 (2019) Winter Peak (19WP)
- Year 2 (2020) Spring (20G)
- Year 2 (2020) Summer Peak (20SP)
- Year 5 (2024) Light (24L)
- Year 5 (2024) Summer Peak (24SP)
- Year 5 (2024) Winter Peak (24WP)
- Year 10 (2029) Summer Peak (29SP)

The higher-queued requests included in this study are listed in **Appendix B**. These requests and associated upgrades were added to the BASE models.

EPE also identified significant deviations³ between the DISIS-2017-001 BASE cases and the MISO DPP reference cases used for the MISO West study. The MISO shoulder case was used for comparison against SPP spring and light load seasons. The MISO summer peak case was used for comparison against SPP summer and winter peak seasons. The identified topology upgrades in the MISO West area were added in the BASE models. The summary of the identified transmission topology changes is provided in **Appendix C**.

Grouping

The interconnection requests listed in **Appendix A** and **Appendix B** are grouped into sixteen (16) active regional groups. The SPP groupings are listed in **Table 4** below. The interconnection requests provided in the interconnection request database have been identified as being potentially impactful to the SPP transmission system due to electrical proximity to SPP facilities.

Table 4: All SPP Groupings

Group #	Area	Group #	Area
1	Woodward, OK	10	Southeast OK/Northeast TX
2	Hitchland, OK	12	Northwest AR
3	Spearville, KS	13	Northwest MO
4	Northwest KS	14	South Central OK
6	South TX Panhandle/New Mexico	15	East SD
7	Southwest OK	16	West ND
8	North OK/South Central KS	17	West SD
9	Nebraska	18	East ND

³ Significant deviations would include additions/removals 161 kV+ facilities, changes to R, X, B, and L > 5% for 161 kV+ facilities, and generation differences > 10 MW for interconnection requests under consideration for inclusion in the analysis.

Development of Base Cases (BC Cases)

The number of base and transfer cases required for each impact study depends on the service requested and fuel type of the study units. **Table 5** outlines the number of cases required per seasonal case for each dispatch scenario. **Table 6** describes SPP Dispatch Criteria used for this analysis.

SPP provided the seven (7) dispatched seasonal power flow cases. The generation online in the BASE models should reflect the dispatch of the ITP study. The two SPP regional groups (Group 9 and Group 15) then had two types of dispatch for their local generation: High-Variable Energy Resource (HVER) and Network Resource Interconnection Service (NRIS). The LVER dispatch was not needed as only renewable requests were under study. The groups and the dispatch resulted in 46 cases with unique dispatches as shown in **Table 5**.

Table 5: SPP Seasons and Cases per Dispatch

Seasonal Case	ERIS HVER	NRIS HVER Cases (G, L)	NRIS HVER Cases (Peak)
+1 Winter Peak (i.e. 19WP)	1 per group	--	1 per study
+1 Spring (i.e. 20G)	1 per group	1 per group	--
+1 Summer Peak (i.e. 20SP)	1 per group	--	1 per study
+5 Light Load (i.e. 24L)	1 per group	1 per group	--
+5 Summer Peak (i.e. 24SP)	1 per group	--	1 per study
+5 Winter Peak (i.e. 24WP)	1 per group	--	1 per study
+10 Summer Peak (i.e. 29SP)	1 per group	--	1 per study
Total Cases	14 Cases	4 Cases	5 Cases
SPP DPP-2017-FEB-West P3	46 Cases (23 BC/23 TC)		

All in-scope higher queued SPP and MISO generators listed in **Appendix B** were added and dispatched per criteria listed in **Table 6**.

For the ERIS cases the SPP region generation offset caused by the queued generators' dispatch was balanced using tier order adjustments. Tier order dispatch was not used in non-SPP regions. The generation offset was adjusted using a uniform scale across all non-NRIS and non-nuclear units in the region. Dispatched cases were solved area interchange, and the system swing generation was kept as close as possible between the BASE case, base case, and transfer case.

For the NRIS cases the SPP region generation offset caused by the queued generators' dispatch was balanced using the Load Ratio Share (LRS) and uniform scale. The LRS determined where generation adjustments were required. The generation offset was sunk using a uniform scale across all non-NRIS and non-nuclear units in each area. LRS was not used in non-SPP regions. The generation offset was adjusted using a uniform scale across all non-NRIS and non-nuclear units in the region. Dispatched cases were solved without area interchange, and the system swing generation was kept as close as possible between the BASE case, base case, and transfer case.

Development of Analysis Cases (Transfer Cases)

All in-scope higher-queued and current study interconnection requests were dispatched as per criteria listed in **Table 6**. For existing SPP interconnection requests included in the scope, if the existing generation dispatch (P_{GEN}) was greater than the expected GI dispatch criteria, the generation was left as-is. If the existing generation dispatch (P_{GEN}) was less than the expected GI dispatch criteria, it was dispatched up to the defined amount.

Generation adjustments are dispatched against Legacy⁴ conventional generation⁵ in the host TO footprint. For the High Variable Energy Resource (HVER) dispatch scenario, all renewable generation facilities are dispatched to 100% within the studied group and at least 20% outside of the study group. Legacy resources and higher-queued conventional units are used to balance generation changes in the HVER scenarios. For the Light Load condition only, renewables are dispatched at or above 0% capacity outside of the study group. Out-group renewables may also be used to balance generation changes in addition to legacy and higher-queued conventional units for this scenario. The HVER dispatch scenario was used with all cases including Winter, Summer, Spring, and Light Load DISIS BASE cases.

For the Low Variable Energy Resource (LVER) dispatch scenario, all conventional generation facilities are dispatched to 100%. The code 00 for this scenario represents that the entire SPP footprint is included as being in-group. Legacy resources are used to balance generation changes. The LVER dispatch scenario is utilized in Winter and Summer DISIS BASE cases, but only used if there is a conventional resource in the current study. There were no conventional resources under study for Feb-17, therefore the LVER dispatch scenario was not used.

For the Network Resource (NR) dispatch scenario, the dispatch levels for the renewable and conventional generation facilities are determined based upon the level of system integration being requested (ERIS and NRIS). For Spring and Light Load, dispatches are group based. For Winter and Summer, the entire SPP footprint is considered “in-group” for the study (like the LVER dispatch scenario). Legacy resources are used to balance generation changes.

For this analysis, MISO’s partial NRIS was accounted for the in NR dispatch, where ERIS only capacity was not dispatched in the NR dispatch scenarios.

Each current study interconnection request was included in the power flow analysis models as an equivalent generator dispatched at the applicable percentage of the requested service amount with 0.95 power factor capability. The facility modeling includes explicit representation of equivalent Generator Step-Up (GSU) and main project transformer(s) with impedance data provided in the interconnection request. All equivalent collector system branches and transmission tie-lines shorter than 20 miles in length are modeled as zero-impedance branches.

Table 6 : SPP Dispatch Criteria

Dispatch Type	Season	Service Type	Variable Energy Resource In Group	Variable Energy Resource Out Group	Conventional In Group	Conventional Out Group
ERIS HVER	All	All	100%	20%*	N/A	N/A
ERIS LVER	Peak	All	20%	20%	100%	100%
NRIS	Spring and Light Load	ERIS	N/A	N/A	N/A	N/A
		NRIS	100%	20%	100%	20%
	Peak	ERIS	N/A	N/A	N/A	N/A
		NRIS	100%	100%	100%	100%
* 0% for DISIS-2017-001 forward ⁶						

⁴ Generators that are found in the SPP footprint in the DISIS BASE cases that do not map to the SPP generator mapping sheet are considered “Legacy” generators.

⁵ Excluding non-adjustable generation such as hydro/run-of-river.

⁶ It should be noted that while the FEB17-West requests are higher queued to the DIS1701, SPP elected to apply this methodology since the requests are lower queued to the DIS1602.

Study Methodology Criteria

Solve Parameters

The following solution parameters were used:

- Fixed slope decoupled Newton-Raphson
- Tap adjustment – stepping
- Switch shunt adjustments – enable all
- Area interchange disabled
- Adjust phase shift
- Adjust DC taps
- VAR limits – apply immediately
- Must solve within five iterations, three or less is preferred

Thermal Overloads

Network constraints are found by using PowerGEM TARA AC Contingency Calculation (ACCC) analysis on the entire cluster grouping dispatched at the various levels.

For Energy Resource Interconnection Service (ERIS):

For ERIS, thermal overloads are determined for system intact (n-0) (greater than or equal to 100% of Rate A - normal) and for contingency (N-1) (greater than or equal to 100% of Rate B – emergency) conditions.

The overloads are then screened to determine which generator interconnection requests have at least

- 3% Distribution Factor (DF) for system intact conditions (n-0),
- 20% DF upon outage based conditions (N-1), or
- 3% DF on contingent elements that resulted in a non-converged solution.

Non-converged contingencies shall also be considered for limited operation service.

For Network Resource Interconnection Service (NRIS):

Interconnection Requests that requested NRIS are also studied in a separate NRIS analysis to determine if any constraint measured greater than or equal to a 3% DF. If so, these constraints are also considered for transmission reinforcement under NRIS.

Contingencies

The contingency set includes all SPP control area branches and ties 69kV and above, first tier Non-SPP control area branches and ties 115 kV and above, any defined contingencies for these control areas, and generation unit outages for the SPP control areas with SPP reserve share program redispatch.

- All branches, ties, shunts, and generators within the following areas:
 - SPP Internal Areas for 65kV – 999kV facilities:
 - 515 – 546, 640, 641, 642, 645, 650, 652, 659
 - SPP External Areas for 100kV – 999kV facilities:
 - 327, 330, 351, 356, 502-504, 600, 615, 620, 627, 635, 672, 680
- NERC, SPP, and Tier 1 Permanent Contingent Flowgates
- SPP T.O. Specific P1, P2, P4, and P5 TPL-004-1 Contingencies
- SPP T.O. Specific Op Guide Implementation

Monitored Facilities

The monitored elements include all SPP control area branches, ties, and buses 69 kV and above, and all first tier Non-SPP control area branches and ties 69 kV and above. NERC Power Transfer Distribution Flowgates for SPP and first tier Non-SPP control areas are monitored. Additional NERC Flowgates are monitored in second tier or greater Non-SPP control areas. Voltage monitoring was performed for SPP control area buses 69 kV and above.

- All branches (thermal)/ buses(voltage) and ties within the following areas:
 - SPP Internal Areas for 65kV – 999kV facilities:
 - 515 – 546, 640 – 659
- NERC, SPP, and Tier 1 Permanent Monitor Flowgates (thermal)

Voltage

For non-converged power flow solutions that are determined to be caused by lack of voltage support, appropriate transmission support will be determined to mitigate the constraint.

After all thermal overload and voltage support mitigations are determined; a full ACCC analysis is then performed to determine voltage constraints. The following voltage performance guidelines are used in accordance with the Transmission Owner local planning criteria.

SPP Areas (69kV+):

Transmission Owner	Voltage Criteria (System Intact)	Voltage Criteria (Contingency)
AEPW	0.95 – 1.05 pu	0.92 – 1.05 pu
GRDA		0.90 – 1.05 pu
KACY		
SWPA		
OKGE		
OMPA		
WFEC		
SWPS		
MIDW		
SUNC		
KCPL		
INDN		
SPRM		

NPPD		
WAPA		
WERE L-V		0.93 – 1.05 pu
WERE H-V		0.95 – 1.05 pu
EMDE L-V		0.90 – 1.05 pu
EMDE H-V		0.92 – 1.05 pu
LES		0.90 – 1.05 pu
OPPD		

SPP Buses with more stringent voltage criteria:

Bus Name/Number	Voltage Criteria (System Intact)	Voltage Criteria (Contingency)
TUCO 230kV 525830	0.925 – 1.05 pu	0.925 – 1.05 pu
Wolf Creek 345kV 532797	0.985 – 1.03 pu	0.985 – 1.03 pu
FCS 646251	1.001 – 1.047 pu	1.001 – 1.047 pu

Affected System Areas (115kV+):

Transmission Owner	Voltage Criteria (System Intact)	Voltage Criteria (Contingency)
AECI	0.95 – 1.05 pu	0.90 – 1.05 pu
EES-EAI		
LAGN		
EES		
AMMO		
CLEC		
LAFA		
LEPA		
XEL		

MP		
SMPA		
GRE		0.90 – 1.10 pu
OTP		0.90 – 1.05 pu
OTP-H (115kV+)	0.97 – 1.05 pu	0.92 – 1.10 pu
ALTW	0.95 – 1.05 pu	0.90 – 1.05 pu
MEC		
MDU		
SPC		0.95 – 1.05 pu
DPC		0.90 – 1.05 pu
ALTE		

The constraints identified through the voltage scan are then screened for the following for each interconnection request.

- 3% DF on the contingent element and
- 2% change in pu voltage

Identification of Network Constraints

ERIS Thermal Non-Converged Constraint Identification and Mitigation

There were no non-converged thermal constraints identified for Group 9 or 15 ERIS interconnection requests.

ERIS Thermal System Intact and Contingency Constraint Identification and Mitigation

There were no system intact or contingency thermal constraints identified for Group 9 or 15 ERIS interconnection requests.

ERIS Voltage Constraint Identification and Mitigation

There were no voltage constraints identified for Group 9 or 15 ERIS interconnection requests.

NRIS Thermal Non-Converged Constraint Identification and Mitigation

There were no NRIS non-converged constraints observed for single contingency (N-1) conditions.

NRIS Thermal System Intact and Contingency Constraint Identification and Mitigation

There were no NRIS thermal constraints identified for single contingency (N-1) and multiple contingency conditions. One NRIS thermal constraint was identified as a contingent network upgrade. **Table 7** below summarizes the contingent NRIS thermal constraint and associated mitigation.

Table 7 : NRIS Contingent Thermal Constraints

Monitored Facility	Mitigation
Sioux City 230 kV Bus-Tie	Sioux City 230 kV Bus-Tie Upgrade

NRIS Voltage Constraint Identification and Mitigation

There were no voltage constraints identified for Groups 9 and 15 NRIS interconnection requests after accounting for network upgrades associated with non-convergence and thermal issues.

Network Upgrade Cost Estimates, Alternative Solutions and Limited Operation Availability

Network Upgrade Cost Estimates

Preliminary cost estimates provided in this analysis are subject to change.

SPP utilizes the one-year-out spring seasonal model for Variable Energy Resources (VERs). The five-year-out summer peak seasonal model is used for conventional fuel type generators. If both fuel types are being studied, both sets of models are utilized. Project distribution factors on the identified upgrades, under system intact conditions, are used to determine cost allocation. The impact each generation interconnection request has on each upgrade project is weighted by the size of each request. Finally, the costs due by each request for a particular project are then determined by allocating the portion of each request's impact over the impact of all affecting requests.

For example, assume that there are three Generation Interconnection requests, X, Y, and Z that are responsible for the costs of Upgrade Project '1'. Given that their respective power transfer distribution factors (PTDF) for the project have been determined, the cost allocation for Generation Interconnection request 'X' for Upgrade Project 1 is found by the following set of steps and formulas:

- Request X, Upgrade Project 1 = $PTDF\%(X) * MW(X) = X1$
- Request Y, Upgrade Project 1 = $PTDF\%(Y) * MW(Y) = Y1$
- Request Z, Upgrade Project 1 = $PTDF\%(Z) * MW(Z) = Z1$

Allocation of Cost for a particular project:

- Request X's Project 1 Cost Allocation (\$) = $\frac{\text{Network Upgrade Project 1 Cost (\$)} * X1}{X1 + Y1 + Z1}$

Repeat previous for each responsible GI request for each Project.

If the current study interconnection request requires a network upgrade for full interconnection service, the study resource will determine the Limited Operation amount available to the request prior to all required network upgrades being in-service. **Table 8** lists the allocated costs for Network Upgrades assigned to current study projects.

Table 8: Network Upgrade Cost Estimates

Interconnection Request	Size	NRIS	Total
J718	45	\$0	\$0
J748	175	\$0	

It should be noted that network upgrades associated with higher-queued projects are also considered as contingent upgrades. These facilities have been included in the models for this study and are assumed to be in service. This list may not be all-inclusive. The interconnection customers, at this time, do not have cost responsibility for these facilities but may later be assigned cost if higher-queued customers terminate their generation.

Limited Operation Availability

The results of the Power Flow identified the system constraints that require mitigation. The Limited Operation Analysis evaluated the most limiting of these constraints for each current study request and identifies an amount of available interconnection service.

Power Flow Analysis results included the thermal overload amount, circuit rating, size, and TDF of each current study request. An initial Limited Operation amount is calculated by identifying the impact of each request on each constraint and identifying a reduced size of each request proportional to the thermal constraint that would result in a circuit loading within the applicable rating. The Limited Operation amount is calculated according to the following equation:

$$\text{Limited Operation amount} = \text{Request MW} - \frac{\text{MVA Rating} * (\text{Overload PU} - 1)}{\text{Request TDF}}$$

With the initial Limited Operation amount request sizes applied to the study cases, ACCC is repeated to verify that the thermal constraints are not observed, or the calculation and verification are repeated until all thermal constraints are mitigated.

Power Flow Analysis results for voltage violations are then further mitigated by identifying the contribution of each request and determination of the required impact reduction is conducted and verified through ACCC to determine the Power Flow Analysis Limited Operation amount for each request.

Limited Operation Results are listed below in **Table 9**. While these results are based on the criteria listed in GIP 8.4.3, the Interconnection Customers may request additional scenarios for Limited Operation based on higher-queued Interconnection Requests not being placed in service.

Table 9: Limited Operation Results

Interconnection Request	Group	Service Type	Available MW Before Mitigation	Most Limiting Constraint
J718	15 E-SD	ERIS	49.98	None
		NRIS	45	None
J748	09 NEB	ERIS	200	None
		NRIS	175	None

Power Flow Analysis

The results of the power flow analysis for interconnection requests under study are embedded in **Table 10**.

Table 10: Power Flow Analysis Results

Results	
Thermal Constraints	 FEB17_Thermal_Constraints.xlsx
Voltage Constraints	N/A
Network Upgrades and Cost Allocation Calculations	 FEB17_Network_Upgrades_Cost_Allocat

Conclusion

A power flow analysis was performed to determine the impact of two (2) MISO GIRs on the SPP transmission system. The results of the Power Flow analysis identified no system constraints that require mitigation. The results of the Power Flow analysis identified one contingent constraint and no additional system constraints required mitigation. Therefore, no DPP-2017-FEB-West generator project has any cost allocation or limited operation availability assigned.

Appendix A: Current Study Interconnection Requests

ERIS

Generation Interconnection Number	Study	Group	Type	G P _{MAX}	SP P _{MAX}	WP P _{MAX}	Service	GEN Area	Point of Interconnection
J718	DPP-2017-FEB	15 E-SD	Solar	49.98	49.98	49.98	ER/NR	XEL	Cherry Grove 69kV Substation
J748	DPP-2017-FEB	09 NEB	Wind	200	200	200	ER/NR	MEC	Highland - Raun 345kV Line Tap

NRIS

Generation Interconnection Number	Study	Group	Type	G P _{MAX}	SP P _{MAX}	WP P _{MAX}	Service	GEN Area	Point of Interconnection
J718	DPP-2017-FEB	15 E-SD	Solar	45	45	45	ER/NR	XEL	Cherry Grove 69kV Substation
J748	DPP-2017-FEB	09 NEB	Wind	175	175	175	ER/NR	MEC	Highland - Raun 345kV Line Tap

Appendix B: Higher Queued Interconnection Requests



Higher_Queued_Pr
jects.xlsx

Appendix C: MISO Topology Updates included in this Analysis

Description	Area Number	kV
Add 2nd 345-161 kV transformer at Sub 39	635	345/161
New Beehive 345/138 kV Substation	356	345/138
Rebuild Messenger - Carbide 69 kV to double circuit	356	69
New Cardinal - Hickory Creek 345 kV	600	138
Rebuild Triboji - Orleans Tap 69kV line	680	69
New Elisha 115/34.5 kV substation using Hubbard transformer	608	115/34.5
Build 115 kV line from Potato Lake tap to Arago distribution substation.	608	115
Construct a new substation in the Hills area.	627	345/69
Construct a new 3 terminal switch station near Lyon	652	69
New 69 kV line to add the new Marion South sub.	627	69
Install 345/138 kV transformer at Gateway	357	345/138
Replace 161 kV circuit breaker at Bondurant Substation	635	161
New 115 kV substation yard adjacent to the Boswell 230 kV yard. A new 230/115 kV transformer will connect both yards.	608	230/115
New +/-75 MVAR STATCOM	608	115
Build a new 115 kV line from the MP Big Rock substation to the GRE Waldo substation	608	115
Extend existing Forbes - Laskin 115 kV Line (38 Line) up to Hoyt Lakes, removing connection to Laskin	608	115
New 115 Tioga Substation	608	115/23
Construct a second 161 kV line from Sub A to Sub 39.	635	161
Reconfigure the 115 kV near Indiana	600	115
Expansion of the existing Hollydale substation to accommodate serving load at Hollydale at 69 kV on a permanent basis.	600, 615	69/13.8
New 115 kV from WAPA's Miles City #2 substation to MDU's new Miles City SW 115 kV substation.	652, 661	115
New 161 kV Bondurant substation	635	161/13
Upgrade Red Rock TR 9 and associated bus work and equipment to higher rating	600	345/115
Rebuild/replace the existing Glenwood 69-13 kV Substation to improve reliability.	680	69/13
Adds 10MVAR cap bank at Turtle Lake	680	69
Rebuild 2 69kV line feeders into new Clear Lake Substation across the street from existing Clear Lake Substation. Install new 69kV breakers and buswork.	600	69
Rebuild and upgrade 14 miles of 69 kV line to 138 kV design (Altoona - Pierson)	635	138
Replace bus tie 3-4 OCB, and pos S OCB at Maline substation	356	138
Upgrade Hazel Creek TR9 to accommodate the interconnection of J460 - Blazing Star 1.	600	345/230
New 345 kV and 230 kV facilities at Big Stone South, Brookings County, and Whapeton to accommodate multiple Generation Interconnection Requests	620	345

Description	Area Number	kV
Add new 69 kV Path (Maquoketa - Delmar)	627	69
Add new 69 kV path (Taft & Brinsdale - Prairie)	620	69
Add parallel branch and transformer at Zachary	356	345/161
Add parallel transformer at Sub 18	635	161/69
Add series capacitor at Briggs Road	600	345
Add series capacitor at Pillsbury	620	230
Reconfigure 69 kV at Court Tap	627	69
Add new 69 kV path at Blomkest	600	69
Reconfigure Adams 161 kV substation	627	161
Add new 69 kV path at Hector Tap	600	69
Add new transformer at Frontier	620	230/115
Add new Raccoon 345 kV station	635	345
Reconfigure 69 kV at Bryantsburg	627	69
Reconfigure 69 kV at Slakes	627	69
Reconfigure 69 kV at Morrell and Cliff Ave	600	69
Reconfigure 69 kV at IPL Tiffin	627	69
Remove transformer at Poplar Lake	680	69
Add new 69 kV path from Ryan to Manchester	627	69
Reconfigure 69 kV at Huntley	627	69
Reconfigure 118.1 kV at Fort Frances	103	118.1
Remove duplicate modeling of transformers at Hancock	627	69
Add Walters 161 tap and 161/69 transformer	627	161
Remove parallel transformer at Henry County	330	161/69
Reconfigure Hall transformer connections	356	138/34.5
Transformer parameter modifications, including all impedance changes >5%	Multiple	Multiple
Branch parameter modifications, including all impedance changes >5%	Multiple	Multiple
Add, remove, and modify various switched shunts	Multiple	Multiple